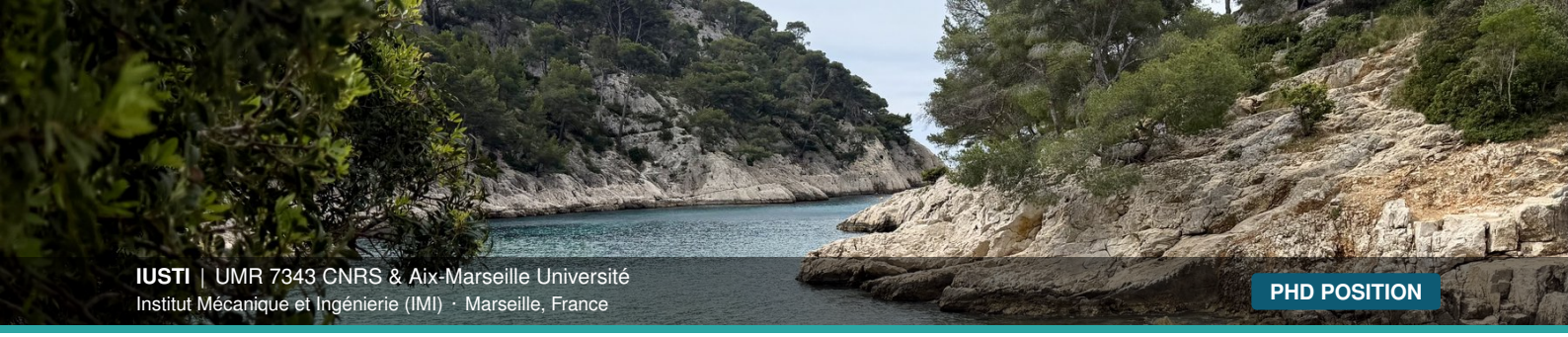


SNSF project *Hierarchy of Scales* · WP 2.1

The crack tip as a rheometer

Measuring nonlinear elasticity in the near-tip fields of soft cracks

* * *

POSITION	PhD student, experimental	**DURATION**	3 years (doctoral contract)	**START**	As soon as possible
PARTNERS	ETH Zurich (D. Kammer, simulations) · CNRS Lyon (M. Adda-Bedia, theory) · IUSTI & IMI, Aix-Marseille Université				

The sub-project. Near the tip of a crack, a soft material is stretched far beyond its linear response: strains become large, the tip blunts, and the fields depart from the square-root singularity of linear elastic fracture mechanics [1–4]. The size and shape of this nonlinear zone therefore encode the constitutive law of the material at precisely the strains that govern failure, strains that are difficult to reach in a conventional rheometer. This PhD project (WP 2.1) will turn this observation into a measurement. By tracking fluorescent tracer particles embedded in hydrogels around propagating cracks [5], the student will reconstruct the full displacement and strain fields near the tip, measure the crack-tip opening displacement (CTOD), and extract the nonlinear elastic response of the material from these fields. Experiments will span different gel chemistries, from highly to weakly cross-linked polyacrylamide, and a range of crack speeds, and will be compared with large-deformation simulations at ETH Zurich and asymptotic theory in Lyon.

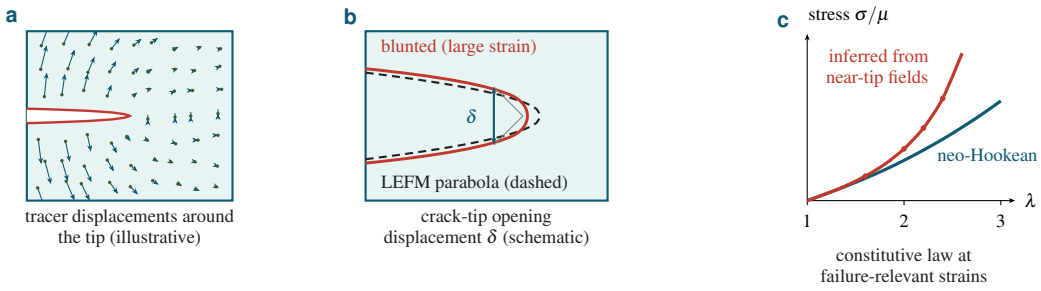


Figure 1. The crack tip as a rheometer (schematic). (a) Fluorescent tracers reveal the displacement field around a propagating crack (arrows: linear elastic Mode I field, for illustration). (b) At large strains the tip blunts and departs from the parabolic profile of linear elastic fracture mechanics, as quantified by the crack-tip opening displacement δ. (c) Near-tip fields yield the stress–stretch response at the strains where the material fails (illustrative strain-stiffening law).

Context. This position is part of the SNSF project "The Hierarchy of Scales in the Fracture of Soft and Slender Bodies" (2025–2029). Whether and how a soft solid breaks depends on several length scales at once: the size of the body itself (its thickness, or slenderness), the nonlinear-elastic zone near the crack tip where the material is stretched far beyond linear response, and the region over which energy is dissipated by molecular processes. In soft materials these scales can be comparable, so classical fracture theories, which rely on their strict separation, break down. The project addresses each scale in a dedicated work package (WP) and connects them through combined experiments, simulations, and theory: experiments at Aix-Marseille Université, simulations in the group of Prof. D. Kammer at ETH Zurich, and analytical theory with Prof. M. Adda-Bedia (CNRS, Lyon). The experimental program is carried out at IUSTI (UMR 7343, CNRS / AMU) in the group of J. Kolinski, equipped with custom tensile loading devices, 3D light-sheet microscopy, and particle-tracking infrastructure, with access to AMU's cryo-electron microscopy facilities. IUSTI and the wider Institut Mécanique et Ingénierie (IMI) have strong expertise in rheology, granular and suspension mechanics, and solid mechanics; collaborations within the lab and across the IMI are strongly encouraged, as are research stays at ETH Zurich. The project will host an international workshop on soft fracture at Monte Verità (Ascona, Switzerland) in November 2027.

Candidate profile. We seek a highly motivated candidate holding, or about to obtain, a Master's degree (M2 or equivalent) in physics, mechanics, materials science, or a related field, with a taste for experiments and quantitative analysis. Experience with particle tracking, digital image correlation, continuum mechanics, or scientific programming (Python or MATLAB) is welcome but not required; training will be provided. The PhD will be carried out on a three-year doctoral contract at Aix-Marseille Université, supervised by J. Kolinski, with day-to-day support from the project's postdoctoral researcher and opportunities for research stays at ETH Zurich. The working language of the group is English.

Contact and application. Send a CV, a one-page statement of interest, and the names of two referees to John Kolinski, john.kolinski@univ-amu.fr • kolinski.science. Applications will be reviewed until the position is filled.

[1] P. H. Geubelle and W. G. Knauss, "Finite strains at the tip of a crack in a sheet of hyperelastic material: I. Homogeneous case," *J. Elasticity* **35**, 61–98 (1994).
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[3] A. Livne, E. Bouchbinder, I. Svetlizky, and J. Fineberg, "The near-tip fields of fast cracks," *Science* **327**, 1359–1363 (2010).
[4] R. Long and C.-Y. Hui, "Crack tip fields in soft elastic solids subjected to large quasi-static deformation — a review," *Extreme Mech. Lett.* **4**, 131–155 (2015).
[5] J. C. Crocker and D. G. Grier, "Methods of digital video microscopy for colloidal studies," *J. Colloid Interface Sci.* **179**, 298–310 (1996).